

Date: June 12, 2026

Project: Autonomous Drone Simulation & Swarm Modeling

Workspace Platform: Ubuntu (WSL2) / Pixi Environment Manager

1. Simulation Environment Setup & Troubleshooting

We successfully configured a fully isolated development environment using the **Pixi** package manager, resolving several toolchain conflicts along the way.

- **ROS 2 & Gazebo Integration:** Resolved dependency pathing issues between rostack-jazzy and conda-forge channels.
- **Missing Toolchain Patches:** * Installed Python build dependencies (kconfiglib, jinja2, empy, pyros-genmsg) via a targeted workspace environment pipeline.
 - Patched missing OS compilation binaries by installing build-essential and cmake natively on the Ubuntu layer.
- **Middleware Implementation:** Compiled the **Micro-XRCE-DDS-Agent** entirely from source and registered its shared libraries using ldconfig to bridge communications between ROS 2 and the flight controller.

2. Firmware Compilation & 3D Vehicle Rendering

- **Firmware Build:** Successfully compiled the complete **PX4 Autopilot (v1.18 alpha flight stack)** for Software-in-the-Loop (px4_sitl) testing.
- **3D Physics Simulation:** Initialized the **Gazebo Harmonic** simulator engine and successfully loaded/rendered a virtual **X500 Quadcopter** model inside a 3D physical workspace.
- **Pipeline Optimization:** Bypassed GUI configuration popups by implementing the **PX4_GZ_STANDALONE=1** launch argument, establishing a seamless, single-command launch process for future sessions.

3. Algorithm Finalization: Slime Mould Algorithm (SMA)

We finalized the selection of the **Slime Mould Algorithm (SMA)** as the mathematical routing and coordination engine for our multi-UAV swarm model.

Core Features of SMA in Swarm Modeling:

- **Dynamic Weighting:** Uses a bio-inspired negative feedback loop to simulate the contraction/expansion of slime mould veins, mapping optimal paths between drones.
- **Exploration vs. Exploitation:** Employs a unique stochastic fitness-weighting wave that helps the drone swarm transition smoothly between spreading out to search an area and converging on a target.

Pros & Cons Analysis

Pros (Advantages)	Cons (Limitations)
High Convergence Speed: Reaches optimal positioning faster than standard Particle Swarm Optimization (PSO) in complex environments.	Premature Convergence Risks: Can occasionally get trapped in local optima if the search space contains highly deceptive, complex obstacles.
High Flexibility: Easily scales mathematically as more quadcopters are added to the swarm partition.	High Parameter Sensitivity: The transition bridge variable ($\$p\$$) requires precise tuning; minor errors can cause the swarm to scatter.
Low Computational Overhead: Doesn't require high-end GPU power, making it ideal for real-time onboard flight computers.	Disconnection Danger: If exploration weights are too high, physical drones might spread beyond communication ranges.

